

FOR FAMILY EYES — TECHNICAL REVIEW

A Technical Conversation

Padel AI Coaching Platform — a conversation between father and son, asking for your brain on the architecture and the moat thesis.

PREPARED FOR

[Son's name]

READING TIME

20–50 minutes

REQUIRED RESPONSE

4 lines on §Q5 and §Q7

AUTHORED

Manoj Maheshwari · 2026-07-01

To: [Son's name] **From:** Papa (Manoj) **Date:** 2026-07-01 **Reading time:** 20 min for the brief + 30 min for the technical depth (if you have it). Honest answers to the questions at the end are the prize.

Why this document exists

You introduced me to Clutch. You play 3x a week. You're on their app. You work at Apple in AI. There is **no one whose technical and product opinion would change my decisions more** on this venture.

I am committing to building an India-first AI padel coaching platform — a Clutch-equivalent but defensible by geography and language, with an expert coach co-founder designing the technique rubric. I have done about 4 hours of due diligence research with Claude. This document is the **distilled output of that research**. I'd like you to:

1. **Read the technical plan below**
2. **Tell me where I'm wrong, where Apple-stack thinking changes the answer, and where the moat thesis breaks**
3. **Not commit any time** — just give direction. I want your brain, not your weekends. Time investment only if we hit commercial traction (paying clubs).

If you accept this lightweight advisory role, the equity ask would be **0.5–1% over 2-year vest as a formal advisor** — designed so you have a real upside if this works, no obligation if it doesn't, and zero conflict with Apple.

Part 1: The product (what we'd build)

A single IP camera per padel court records every match. Cloud AI runs computer vision: player tracking, ball trajectory, pose estimation, court keypoints, and most importantly — **shot classification scored against an expert padel coach's rubric**. Players get a WhatsApp message 10 min after every match: "Your bandeja: 3.2/5, your hip rotation is 18° below ideal. Drill — shadow swing with hip cue, 5 min/day."

The MVP scope: - 1 camera per court (Reolink RLC-823A 4K, PoE, ~₹12k) - Edge buffer (Pi 5 8GB + 1TB NVMe) for offline reliability - RTSP stream → cloud GPU (RunPod RTX 3090) → analysis - Player PWA delivered via WhatsApp link - Club dashboard for court owners

The product roadmap: - V1 (Month 3): Movement + shot histogram + skill score - V2 (Month 6): Shot quality scoring + drill recommendations + clips - V3 (Month 12): Cross-club player identity ("Skill Score") + matchmaking - V4 (Year 2): Edge inference (this is where I want your Apple Neural Engine view)

Part 2: The technical stack we propose

2.1 OSS we'd fork as the base

COMPONENT	OSS PICK	WHY
Full pipeline scaffold	Joao-M-Silva/padel_analytics	Player tracking + pose + ball + court keypoints already wired. Author actively seeking collaborators.
Ball tracking	yastrebksv/TrackNet	Cleanest TrackNet implementation. Padel-retrain candidate.
Court keypoints	yastrebksv/TennisCourtDetector	Retrain on padel court keypoints.
Reference pipeline	ArtLabss/tennis-tracking	Tennis HawkEye OSS — most complete reference.
Training datasets	UPC-ViRVIG/PadelVic	Padel-labeled video datasets.

2.2 Live architecture (proposed)



2.3 The hard ML problem (where you can save us months)

The classifier is the moat. Pipeline diagram shows step 5 ("shot classifier"). This is the one piece **not solved in OSS**. We have to train it.

Plan: - **Input:** Per-frame pose-keypoint sequence (~30 frames at 30fps = 1 sec around ball impact) + court-relative position + ball trajectory feature - **Output:** Shot class (bandeja, vibora, smash, volley, bajada, chiquita, x3, serve, return, lob, drive) + **quality score 1-5** + **specific deviations from ideal**

form - Training data: Coach designs rubric (10-15 page IP doc) + labels 100 gold-standard clips + 2 trained labellers do bulk labelling (target 500-700 clips for V1) - **Architecture (placeholder):** Small MLP or temporal CNN on pose-keypoint features → multi-head (class + quality + deviation flags) - **Target accuracy V1:** 70% top-1 class accuracy

The big technical question I want your view on: Is this approach right? Or should we be looking at: - (a) Temporal transformer over keypoints? - (b) Fine-tune a pretrained pose-classifier (e.g., MotionBERT)? - (c) Multi-modal (pose + ball trajectory features cross-attention)? - (d) Self-supervised pretraining on unlabeled padel video first, then small labeled fine-tune?

2.4 Hardware bill of materials (per court)

ITEM	SPEC	INR
Camera	Reolink RLC-823A 16x or Hikvision DS-2CD2T87G2	12,000-22,000
Edge buffer	Pi 5 8GB + 1TB NVMe + case + cooling	15,000
PoE switch	TP-Link TL-SG1005P	3,000
Cabling + mount + IP66 box	Local	2,000
Per-court total		~₹32,000-42,000

2.5 Cloud cost per court per month (at 120 matches/court/month)

ITEM	INR
RunPod GPU inference (~3 min/match × ₹2.5/min)	360
R2 storage + egress	150
Supabase free/pro tier	30
WhatsApp delivery (own infra)	30
Total COGS / court / mo	~₹540

At ₹5,000/court/mo ASP → **~89% gross margin.**

Part 3: Edge inference (V2/V3) — the Apple-stack question

This is where you have a 10x view I don't.

Year-2 plan currently has us migrating from “cloud-only inference” to “edge inference on a small device per court” — primary motivations: 1. Cost: cloud inference at scale (1,000+ clubs) gets expensive 2. Latency: live in-match insights need <1 sec round trip 3. Bandwidth: 4K RTSP to cloud at scale = ISP grief at clubs

Default plan picks **NVIDIA Jetson Orin Nano (₹40k) or Jetson Orin NX 16GB (₹70k)** as the edge device. That's the safe path.

The Apple-stack alternative I want your view on: - Mac mini M4 (₹60–70k) as the edge box per court, running CoreML-compiled models on Neural Engine - Pros: cheaper than Orin NX, Apple Neural Engine is genuinely top-tier for vision workloads, you'd know how to optimize this in your sleep - Cons: macOS is not a server OS most people use for production deploys; software ecosystem (RTSP capture, ffmpeg, headless service management) is slightly more awkward - **Reality check question:** Have you seen anyone do production CV pipelines on Mac mini in a multi-site deploy? Or is this a thought experiment? If real, can you point me at any case studies?

There's a Mac Pro M2 Ultra angle for a regional inference node serving 10–20 clubs from a colocated rack. Same question.

Part 4: Competitive landscape (what we found)

You know Clutch. There are 7–8 European competitors (Clutch, Wingfield, Padelytics, Eyes-On, GAMETRAQ, SPASH, PlaySight). The interesting find this week:

Epic Padel (\$10M seed Sept 2025) — US-based rollup that OWNS Clutch AI + Padel India (chain of courts) + Red Padel (the world rating) + multiple US club chains. They're launching **Zero.40 in 2026** — “club operating system” combining all of this. Clutch India is their Bangalore arm — Curved Consistency Technologies Pvt Ltd, co-founded by **Sharath Kumar** (ex-Ninjacart, ex-BharatPe).

China that I missed in the first pass: - (Xiaoqiuquan) — ex-ByteDance + ex-DJI founders, AI coach for tennis/badminton/table-tennis/pickleball, already top-50 on Chinese sports app charts, ~30% MoM user growth. **They will add padel within 12 months.** - Lumistar (Shenzhen) — AI tennis training systems (TERO launched Mar 2026) - SenseTime — already runs basketball AI analytics for Chinese pro teams - Hikvision/Dahua — they manufacture the cameras we'd buy; could vertically integrate

Our moat thesis (please critique honestly): - ❌ Tech moat — we agree there isn't one; it's commodity ML - ✅ India geopolitical wall vs Chinese SaaS (TikTok/PUBG-style) - ✅ Federation distribution lock (Asian Games 2026 catalysed national federations forming NOW) - ✅ Padel coach co-founder = proprietary rubric dataset - ✅ WhatsApp-native distribution + Hindi/Arabic UX depth

The biggest risk I see: if Sharath Kumar's team (Epic Padel-funded) moves fast in Mumbai/Bangalore before we get 25 clubs locked, the geographic moat thesis erodes. That's a 6-month race, not a 24-month one.

Part 5: Honest open questions for you

You can answer these inline below each one, save the file, and send back. Or schedule 30 min and we talk.

Q1. Clutch as a user

Have you used Clutch enough to tell me **what is genuinely good** about the product and **what is weak**? What would make you switch to a different product if it appeared in your club? What do you wish it did that it doesn't?

Your answer:

Q2. The classifier approach

Of the four options I listed in §2.3 (MLP, MotionBERT, multi-modal cross-attention, self-supervised pretraining), which would you bet on for V1? Why? What would you change about the framing of the problem?

Your answer:

Q3. Edge inference path

Is Mac mini M4 + CoreML a real production CV deploy path in 2026? Or do we go safely with Jetson Orin? If you'd pick Apple stack, what's the architecture you'd build?

Your answer:

Q4. Apple internal angles

Without revealing anything confidential — are there public Apple frameworks (Vision, ARKit body pose, CoreML model zoo) that we should be paying attention to? Anything from VisionPro Body Pose APIs that would shortcut our pose-keypoint work?

Your answer:

Q5. The moat

Honest critique: does the geopolitical + federation + coach + WhatsApp moat actually hold up under pressure, or am I rationalizing? What would you ADD to the moat thesis if you were investing?

Your answer:

Q6. The build vs partner question

There's a real option: **don't compete with Clutch — try to become their India partner instead**. Sharath Kumar is in Bangalore, the company is Indian-incorporated, they likely need install/ops help in Mumbai/Delhi. If we did this, my AX distribution muscle goes into THEIR product, we get revenue share, no fundraise pressure. Less glory, less risk. **What's your take?**

Your answer:

Q7. Your honest "should I do this" view

You're the closest person to me who plays this sport AND understands the tech. **Should I do this venture?** Or is it a distraction from AutomationXpert and I should send the LinkedIn DM to Sharath to find partnership / advisor / nothing? No need to be polite.

Your answer:

Q8. Anything I missed

What would you have asked that I didn't?

Your answer:

Q9. The advisory role

If yes — 0.5–1% advisor equity over 2-yr vest, no time commitment, monthly 30-min sync, you respond when asked specific questions (like in this doc). Acceptable structure? Want to change anything?

Your answer:

Part 6: Reference reading (only if you want depth)

- **Strategy:** `~/Documents/padel-clone/PADEL_AI_VENTURE_PLAN.md` — full venture plan v0.2
- **Execution:** `~/Documents/padel-clone/PROJECT_REPORT.md` — operational plan with 30-day punch list
- **Technical feasibility:** `~/Documents/padel-clone/TECHNICAL_FEASIBILITY_REPORT.md` — formal feasibility report
- **OSS we'd fork:** https://github.com/Joao-M-Silva/padel_analytics (the base) + <https://github.com/yastrebksv/TrackNet> (ball)

Thank you for reading this far. Even a 20-min skim and 4-line response on the moat (§Q5) and “should I do this” (§Q7) would be incredibly useful. No pressure, no obligation — just your brain. Love, Papa.

Companion document set: Strategic Venture Plan · Project Report · Technical Feasibility · Commercial Report · Base Club Offer · Technical Review Request.

Single source of truth maintained in `~/Documents/padel-clone/`.